Title : Project Registration & Progress Review **FF No. 180**

Department: DESH	**Academic Year: 2024-25**
Semester : 1	**Group No. : 2**
Project Title: Raspberry Pi 4-Based Golf Ball Tracking and Drop Location Prediction Using Computer Vision	
Project Area: Embedded Systems, Computer Vision, Artificial Intelligence, Internet of Things	

Group Members Details:

Sr. No.	Class & Div.	Roll No.	P.R.No.	Name of Student	Contact No.	Email ID
1	AIML-F	7	12410008	Shoeib Abrar	9322802005	shoeib.abrar24@vit.edu
2	AIML-F	8	12414275	Shubhankar Parsewar	8080578728	shubhankar.parsewar24@vit.edu
3	AIML-F	9	12414256	Swanandi Shukla	9960409976	swanandi.shukla24@vit.edu
4	AIML-F	10	12414273	Siddhesh Amrutkar	9158454466	siddhesh.amrutkar24@vit.edu
5	AIML-F	11	12411235	Apurva Sidhawar	9112245390	apurva.sidhawar24@vit.edu
	AIML-F	12	12411766	Omkar Singare	8888018738	omkar.singare24@vit.edu

Name of Internal Guide: Prof. Jyoti Khubchandani
Contact No. & Email ID: +91 8208166897, jyoti.khubchandani@vit.edu

Project approved / Not approved

Guide **Project Coordinator** **Head of Department**

FF No 180

Project Synopsis

IMPORTANCE OF THE DECIDED PROJECT:

In golf, a considerable amount of gameplay time is lost in locating the ball after a shot. This project aims to solve that problem using a Raspberry Pi 4-based single-camera system that leverages computer vision and physics modeling to track the golf ball after impact and predict its landing location. The system factors in the wind speed, shot angle, impact force, and goal hole position. It is designed to be compact, cost-effective, and deployable on any golf course to assist players and organizers by reducing game time and enhancing overall efficiency.

STEPS TO DO THE PROJECT / METHODOLOGY:

1. Hardware Setup
   - Use Raspberry Pi 4 with Pi Camera or high-speed USB webcam.
   - Optional wind sensor module interfaced via GPIO.

2. Camera Calibration and Input Feed
   - Mount camera at a fixed position overlooking tee area.
   - Calibrate camera for distance and scale.

3. Club and Ball Detection
   - Use OpenCV to track motion of the golf club and detect impact with the ball.
   - Estimate launch parameters (velocity, angle) using frame analysis.

4. Trajectory and Physics Modeling
   - Use kinematic equations and wind data to simulate the ball's flight path.
   - Estimate drag and lift coefficients based on historical data or regression.

5. Landing Zone Prediction
   - Predict and visualize the approximate drop location of the ball.
   - Overlay prediction on a top-down course map or image.

6. Optimization and Testing
   - Run multiple trials to validate predictions.
   - Optimize performance for real-time operation on Raspberry Pi 4.

TENTATIVE EXPECTED RESULT FROM THE PROJECT:

The system is expected to accurately track the golf ball using a single-camera setup and estimate its drop location in real-time using Raspberry Pi 4. By incorporating wind speed and analyzing the club’s impact, it provides a close prediction of the landing spot, reducing search time and improving gameplay flow. This can serve as a foundation for smart sports systems in outdoor environments with minimal infrastructure.

Group No.	2		
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1 **Group No.: 2** **Date: 28 April 2025**

Progress Review Report

1. Summary of Progress:

- The literature review and requirement analysis have been successfully completed. Relevant research papers and existing models were studied to define the project scope and technical requirements.
- The preliminary design of the detection and prediction model has been developed. Initial frameworks and algorithms have been outlined, setting a foundation for further development.
- Wind factor integration is currently in progress. Work is underway to incorporate wind-related parameters into the model to enhance prediction accuracy.

2. Next Steps:

- Complete the integration of wind factor variables.
- Begin detailed implementation of the detection and prediction model.
- Conduct initial testing and validation.

3. Challenges:

- Adjusting model parameters to account for dynamic wind conditions without compromising prediction stability.
- Ensuring seamless integration between environmental factors and model output.

Signature of Guide:

Review No.: 2 **Group No.:** **Date:**

Progress Review Report

IoT–ML Pipeline for Golf Ball Trajectory and Drop Forecasting

Overview:

The project has made substantial progress with the successful setup of the **Raspberry Pi 4** and the development of a fully functional **golf ball tracking and drop prediction system**. The system performs real-time ball detection, and estimates drop location on a scaled golf course map. Key achievements include smooth integration of YOLOv8 and HoughCircles, and reliable frame processing on edge hardware.

Faced Challenges During Development:

1. Processing high-speed frames in headless mode
2. Reducing false positives from non-spherical objects
3. Mapping trajectory to real-world distance on a 2D image

Solutions Implemented:

1. Integrated libcamera-vid and subprocess for headless capture
2. Applied Hough Circle Transform to confirm spherical detections
3. Used scaling logic in marker.py for accurate drop projection

Outcome:

1. The system now detects golf balls in real time (~60 FPS)
2. Drop location is predicted with an MAE of ~2.43 meters
3. Operates fully offline on Raspberry Pi 4, with minimal resource load

Next Steps:

1. Develop optional GUI/dashboard or web interface for results visualization

Conclusion:

With core detection and projection features implemented and optimized for Raspberry Pi 4, the project is entering its final refinement phase. The system is technically robust, reliable, and progressing on track toward completion.

Signature of Guide: